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1

COMP

Neuromuscular junction

You should be able to: Trace transmission at the neuromuscular junction from the motor neuron action potential to the end plate potential and explain why every action potential in the motor neuron produces one in the muscle fiber.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw the axon terminal, the cleft, and the motor end plate large enough to number six steps in order: action potential arrives, calcium channels open, vesicles fuse, ACh crosses, receptor channels open, end plate potential. Mark which ion moves at each channel and which direction. Then write one line: the end plate potential is far bigger than it needs to be, so say what that margin buys the muscle.

- ☐ **B** Draw the same junction three times, normal, then with the receptors on the end plate blocked, then with the calcium channels in the terminal blocked. In each version show how far the signal gets before it dies and circle the last structure still working. One line: two of the three look identical to someone watching the patient, so write what you would measure to tell those two apart.

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2

COMP

Excitation contraction coupling

You should be able to: Trace excitation contraction coupling from the T tubule through the DHP and ryanodine receptors to calcium release from the sarcoplasmic reticulum.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw one T tubule diving into the fiber with a terminal cisterna of sarcoplasmic reticulum on each side. Put the DHP receptor in the T tubule membrane and the ryanodine receptor in the SR membrane, touching each other. Draw the action potential arriving and use arrows to show what moves and what only changes shape. Mark where the calcium comes from. One line: say why the T tubule has to go inside the fiber rather than stay on the surface.

- ☐ **B** Draw skeletal and cardiac coupling side by side with the same structures in both. In the skeletal one, show the DHP receptor opening the ryanodine receptor by physical contact. In the cardiac one, show calcium entering from outside first and that calcium opening the ryanodine receptor. Label the calcium source in each. One line: one of the two stops working if you remove calcium from the fluid outside the cell, so name it and say why.

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3

COMP

Crossbridge cycle

You should be able to: Order the steps of the crossbridge cycle, binding, power stroke, detachment, and reactivation, and state where ATP binds and where it is hydrolyzed.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw the cycle as four myosin heads around a loop in order: binding, power stroke, detachment, reactivation. At each head show the thin filament position and what is sitting on the head, nothing, ATP, or ADP and phosphate. Put a star where ATP binds and a different mark where ATP is hydrolyzed, because they are not the same step. One line: name the step that stalls in rigor mortis and what the head is stuck to.

- ☐ **B** Draw a stretch of thin filament with a ruler under it, and draw the myosin head at all four points of the cycle above it, showing how far the filament has traveled by each point. Mark the one step where movement actually happens and the three where the filament does not move. Then write one line on why taking ATP away leaves the muscle stiff rather than limp, using the step you marked.

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4

COMP

Calcium and the regulatory proteins

You should be able to: Explain how calcium binding to troponin moves tropomyosin off the myosin binding site and predict what happens to contraction if calcium cannot be released.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw a stretch of thin filament twice, with actin, tropomyosin, and the troponin complex on both. In the first, no calcium, with tropomyosin lying over the myosin binding sites. In the second, calcium bound to troponin and tropomyosin rolled away. Draw a myosin head reaching in on both and show what it meets. One line: calcium never touches actin or myosin here, so name what it does bind and what that binding moves.

- ☐ **B** Draw a resting sarcomere with a full sarcoplasmic reticulum, then draw the same sarcomere in a fiber whose ryanodine receptors will not open. In both, show the calcium, the troponin, the tropomyosin, and the myosin heads. The nerve, the end plate potential, and the ATP supply are all normal in the second one. One line: say whether that muscle feels stiff or limp to a doctor moving the limb, and why.

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5

COMP

Relaxation and calcium removal

You should be able to: Explain how SERCA and acetylcholinesterase end a contraction and predict the result if either one fails.

PICK ONE AND DO THAT ONE

☐ **A** Draw both shut off jobs on one page: acetylcholinesterase chopping ACh in the synaptic cleft, and SERCA in the sarcoplasmic reticulum membrane pumping calcium back in. Mark which of the two costs ATP. Under each, draw a small graph of cytosolic calcium against time and shade the part of the falling limb that job is responsible for. One line: say why relaxation burns energy when it looks like nothing is happening.

☐ **B** Draw two failures side by side, acetylcholinesterase blocked by an organophosphate, and SERCA stopped. For each, show where the signal is stuck, ACh piled up in the cleft or calcium piled up in the cytosol, and draw the muscle underneath in the state it ends up in. One line: both muscles are contracted and will not release, so name the one thing you could measure or do that separates the two.

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6

COMP

The muscle twitch

You should be able to: Label the latent period, contraction phase, and relaxation phase on a twitch tracing and explain what limits each phase.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw a twitch tracing with time on the x axis and tension on the y, and put an arrow on the axis at the stimulus. Bracket and label the latent period, the contraction phase, and the relaxation phase. Beside each bracket write the one event inside the fiber that sets how long that phase lasts. One line: the tension trace is flat through the latent period, so say what is going on during it.

- ☐ **B** Draw three traces stacked on one shared time axis: the muscle action potential on top, cytosolic calcium in the middle, and tension at the bottom, all lined up. Show that the action potential has finished long before tension peaks. One line: say which of the upper two curves the tension curve actually follows, and what that lag lets you do by stimulating again.

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7

COMP

Summation and tetanus

You should be able to: Explain wave summation and distinguish unfused from fused tetanus by stimulus frequency and calcium accumulation.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw one time axis with stimulus marks below it, and above it draw four tension traces at rising frequency: a single twitch, two twitches close enough to add, unfused tetanus, and fused tetanus. Keep the y axis identical on all four so the heights are comparable. Mark where the trace dips between stimuli and where the dips disappear. One line: say what stops dipping first, the tension or the cytosolic calcium.

- ☐ **B** Draw cytosolic calcium against time for a low frequency train and a high frequency train, one above the other, with SERCA pulling calcium back between stimuli in both. Show why calcium stacks up in one and clears in the other. Under each, draw the tension trace it produces. One line: the action potentials themselves never add together, so write what does.

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8

COMP

Motor units and recruitment

You should be able to: Define the motor unit and explain how size of the unit and orderly recruitment grade the force a whole muscle produces.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw one alpha motor neuron with its axon branching onto every fiber it owns, then draw a much larger motor unit beside it. Name a real muscle for each and label which is which. Underneath, draw a force trace as units are switched on one at a time, smallest first. One line: say why an eye muscle and a calf muscle are built with different unit sizes, in terms of the job each one has.

- ☐ **B** Draw a rising force curve for a whole muscle, from a light hold up to maximum, with force on the y axis and effort on the x. Along the curve, mark the stretch where new motor units are still being added and the stretch where the units already firing are simply firing faster. At three points along it, draw the fiber type being brought in. One line: name the fibers that lifted the coffee cup, and say why those and not the others.

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9

COMP

Length tension relationship

You should be able to: Interpret a length tension curve and explain why tension falls at very short and very long sarcomere lengths.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw the length tension curve with sarcomere length on the x axis and active tension on the y, and mark the plateau at about 2.0 to 2.2 micrometers. Above three points on the curve, one short, one on the plateau, one long, draw the sarcomere at that length with the thick and thin filaments overlapping correctly. One line: tension falls at both ends of this curve, so say why the two ends fall for different reasons.

- ☐ **B** Draw a sarcomere at a very short length and at a very long length, big enough to show the Z discs, the thick filaments, and where the thin filaments end. In the short one, show the two things that physically get in the way. In the long one, show what has run out. One line: say which of the two you can actually reach in a joint in your own body, and what stops you reaching the other.

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10

COMP

Isometric and isotonic contraction

You should be able to: Distinguish isometric, concentric, and eccentric contractions and predict shortening velocity from the load applied.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw the same arm and dumbbell three times: holding it still, curling it up, and lowering it under control. Label each isometric, concentric, or eccentric, and draw an arrow for whether the active muscle is staying the same length, shortening, or lengthening. Beside each, write whether the load is smaller than, equal to, or larger than the tension the muscle is making. One line: say which of the three leaves you sorest two days later.

- ☐ **B** Draw a force velocity curve with load on the x axis and shortening velocity on the y. Mark the point where velocity is highest and the point where velocity hits zero, and write what the load equals at that second point. Beside the curve, draw the arm at three loads along it. One line: the muscle is fully activated at every point on this curve, so say what is actually changing as the load climbs.

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11

COMP

Muscle fiber types

You should be able to: Compare slow oxidative, fast oxidative glycolytic, and fast glycolytic fibers by speed, ATP source, fatigue resistance, and typical task.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw three fibers in cross section at the same scale, slow oxidative, fast oxidative glycolytic, and fast glycolytic. Draw the differences you could actually see: fiber diameter, mitochondria, capillaries wrapped around it, and myoglobin shading. Under each, write its main ATP source, its fatigue resistance, and one task it does. One line: name the single feature on your drawing that best predicts fatigue resistance, and defend picking that one.

- ☐ **B** Pick a sport or a job, not marathon running, and draw the muscle that person needs as a bundle in cross section with the fiber types shaded in the proportions you expect. Beside it, draw a bundle for someone doing the opposite kind of work. Under both, draw force against time out to ten minutes. One line: say where your two curves cross and what that crossing point means for choosing between the two athletes.

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12

COMP

ATP sources during activity

You should be able to: Order creatine phosphate, anaerobic glycolysis, and oxidative phosphorylation by how quickly each supplies ATP and how long each can sustain it.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw one time axis running from a few seconds out to an hour, and draw three curves above it for the rate of ATP supply from stored creatine phosphate, from anaerobic glycolysis, and from oxidative phosphorylation. Show where each peaks and where each fades out. Label which of the three needs oxygen. One line: the fastest source is also the shortest lived, so write what a sprinter has already used up by the ten second mark.

- ☐ **B** Draw three fuel tanks in a row, each sized to how long it can run the muscle, and put a tap on each sized to how fast it delivers. Keep the tank size and the tap size clearly separate, because they are not the same property. Draw the order the muscle opens them in. One line: name the biggest tank and say why the muscle still cannot start a sprint on it.

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13

COMP

Fatigue and oxygen debt

You should be able to: Distinguish central from peripheral fatigue and explain the metabolic basis of excess post exercise oxygen consumption.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw the path from motor cortex to crossbridge as a row of boxes: brain, spinal cord, motor neuron, neuromuscular junction, T tubule, sarcoplasmic reticulum, crossbridge. Draw a vertical line through the row separating central from peripheral. Put a cross at two plausible failure points on each side of your line. One line: name the test you would run on a tired person to tell you which side of that line has failed.

- ☐ **B** Draw oxygen consumption against time across three stages, rest, a hard bout of exercise, then recovery, and shade the region after exercise where consumption is still above resting. Inside that shaded region, draw the things being restored or paid for, one small item each. One line: the muscle stopped working minutes ago, so say why oxygen use has not come back to baseline yet.

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14

COMP

Muscle plasticity

You should be able to: Predict the change in fiber size, capillary supply, and mitochondrial content produced by endurance training, resistance training, and disuse.

PICK ONE AND DO THAT ONE

☐ **A** Draw one fiber in cross section as your baseline, then draw three more at the same scale beside it: after months of endurance training, after months of heavy resistance training, and after six weeks in a cast. In each, draw mitochondria, capillaries, and myofibrils, and keep the diameters honest. One line: say which of the four could lift the most and which could go the longest, and why those are not the same fiber.

☐ **B** Draw three arrows leaving one starting fiber, labeled endurance training, resistance training, and disuse. Along each arrow, draw what changes early and what changes only after months. Then draw what happens on each path when the stimulus stops. One line: one of these adaptations is lost far faster than it was gained, so name it and say what that means for someone coming back from an injury.

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15

COMP

Electromyography and grip fatigue

You should be able to: Record grip force and surface EMG over a sustained contraction and relate the decline in force to recruitment and fatigue.

PICK ONE AND DO THAT ONE

- ☐ **A** Before you squeeze anything, draw the two traces you predict for a long hold at a constant submaximal grip force: force on top, EMG amplitude below, both against time. Then run it and draw the real traces on the same axes in a second color. Mark where prediction and data part company. One line: the force was held flat on purpose, so say what the EMG had to do to keep it there and what that tells you about the units doing the work.

- ☐ **B** Draw the raw EMG burst from the start of a maximal squeeze and from the end of it, side by side at the same scale, with the grip force trace running underneath across the whole hold. Mark on the force trace the moment you first felt it slipping. One line: say whether the falling force proves the motor neurons stopped firing, and name what else could produce the same trace.

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16

COMP

Twitch and tetanus simulation

You should be able to: Generate twitch and tetanus tracings in a muscle simulation by varying stimulus voltage and frequency and identify threshold and maximal stimulus.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw peak twitch tension against stimulus voltage, from zero up past the point where the curve flattens. Put arrows on the x axis at threshold and at maximal stimulus. Above three voltages along the curve, draw the twitch tracing you would get there. One line: nothing about any single fiber changed as you turned the voltage up, so write what did change to make the whole muscle pull harder.

- ☐ **B** Draw a grid with stimulus voltage on one axis and frequency on the other, and sketch the tracing you would get at four corners of it: low voltage low frequency, low voltage high frequency, maximal voltage low frequency, maximal voltage high frequency. Label each tracing twitch, tetanus, or something in between. One line: say which knob you turn for the strongest contraction available, and why the other knob on its own cannot get you there.

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17

COMP

Motor control hierarchy

You should be able to: Order the levels of motor control from spinal reflex through brainstem to cortex and state what each level contributes to a voluntary movement.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw the levels stacked, spinal cord at the bottom, brainstem in the middle, cortex on top, and trace one voluntary movement of your own choosing, not walking, from the decision down to the muscle. At each level write what that level contributed. Draw the cerebellum and basal ganglia off to the side with arrows showing they adjust rather than command. One line: remove the top level only, and say what movement the person can still produce.

- [ ] **B** Draw the same three levels and drop a hot pan into the picture. Trace the withdrawal reflex through the lowest level alone, then trace the slower conscious response that arrives afterwards through all three. Put both on a shared time axis underneath. One line: the reflex is faster because it skipped something, so name what it skipped and what the person gave up by skipping it.

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18
COMP

Corticospinal pathway

You should be able to: Trace the corticospinal pathway from the primary motor cortex to the skeletal muscle and identify where it crosses.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw a coronal outline of the brain, brainstem, spinal cord, and one arm. Run the corticospinal neuron from the precentral gyrus down through the internal capsule, midbrain, and pons, and mark the pyramidal decussation in the medulla where most fibers cross. Synapse it in the ventral horn on the lower motor neuron and continue out to the muscle. Label upper and lower motor neuron. One line: name the side of the body the left motor cortex moves, and the structure that decided that.

- ☐ **B** Draw the pathway once, then place a lesion on it at three sites: in the motor cortex, in the medulla just above the crossing, and in the spinal cord below the crossing. For each, shade the side of the body that ends up weak. One line: two of your three lesions weaken the same side, so write what a clinician would look for to separate those two.

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19

COMP

Cerebellum and basal ganglia

You should be able to: Contrast the contributions of the cerebellum and the basal ganglia to movement and match a described motor sign to the structure involved.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw a hand reaching toward a target three times: normally, with cerebellar damage, and with basal ganglia damage. Draw the actual path the hand takes each time rather than writing about it. Beside each path, draw the tremor if there is one and mark the point in the movement where it shows up. One line: both patients tremble, so say when each one's tremor appears and what that timing tells you about the structure involved.

- ☐ **B** Draw a loop diagram with the motor cortex at the top, the cerebellum on one side and the basal ganglia on the other, each with its arrows out to and back from the cortex. Label what each loop does to the command, comparing it against feedback or selecting and scaling it. One line: name the loop that needs sensory information from the moving limb to work, and say what happens to it when that information is lost.

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20

COMP

Upper and lower motor neuron signs

You should be able to: Distinguish upper from lower motor neuron lesions by tone, reflexes, and muscle bulk and assign a set of findings to the correct level.

PICK ONE AND DO THAT ONE

- ☐ **A** Draw the two neuron chain from cortex to muscle once, and mark two lesion sites, one on the upper neuron and one on the lower. Beside each, draw the limb as it behaves: the tone, the reflex hammer response drawn as a big or a small kick, the muscle bulk, and whether the muscle twitches on its own at rest. One line: both patients are weak, so name the single finding you would check first and say why that one.

- ☐ **B** You are handed a limb with raised tone, brisk reflexes, normal bulk, and a big toe that goes up when you stroke the sole. Draw the neuron chain and mark where the lesion has to be. Then draw a second chain with the opposite findings and mark that lesion. Label both. One line: one of these two patients wastes away quickly and the other does not, so say what that muscle lost contact with.

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